MECHANICAL PROPERTIES OF SOLIDS & FLUIDS

FORMULA SHEET | JEE MAINS | Let's Play With Physics

1. STRESS, STRAIN & HOOKE'S LAW

$$\text{Stress} = \frac{F}{A}\ [\text{N/m}^2], \qquad \text{Strain} = \frac{\Delta L}{L},\ \frac{\Delta V}{V},\ \theta$$

Hooke's law: Stress $= E \times$ Strain (within elastic limit)

Modulus	Formula	For
Young's Y	$\frac{FL}{A\,\Delta L}$	solids (wires)
Bulk B	$-\frac{\Delta P}{\Delta V/V}$	solids, liquids, gases
Shear (rigidity) η or G	$\frac{F}{A\,\theta}$	solids only

Compressibility $K = \frac{1}{B}$

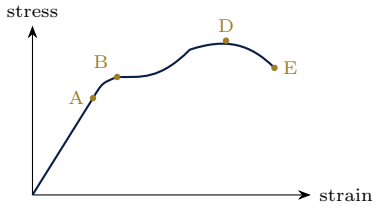
Poisson's ratio: $\sigma = \dfrac{\text{lateral strain}}{\text{longitudinal strain}} = -\dfrac{\Delta d/d}{\Delta L/L}$, $-1 < \sigma < 0.5$

$$Y = 3B(1-2\sigma) = 2\eta(1+\sigma), \qquad \frac{9}{Y} = \frac{1}{B} + \frac{3}{\eta}$$

⋆ Common Pitfall

- Modulus depends on material only, not on length or area
- Steel is more elastic than rubber ($Y_{\text{steel}} > Y_{\text{rubber}}$)
- Liquids & gases have only bulk modulus; $\eta = 0$ for fluids

2. STRESS–STRAIN CURVE



A: proportional limit; B: yield point; D: ultimate (tensile) strength; E: fracture

- Ductile: large plastic region (Cu, Al); Brittle: fracture soon after B (glass)
- Elastomers (rubber): no proportional region, large elastic strain

3. ELASTIC ENERGY & SPECIAL CASES

$$U = \tfrac{1}{2}F\Delta L = \tfrac{1}{2}\frac{YA}{L}(\Delta L)^2$$

$$u = \frac{U}{V} = \tfrac{1}{2}\,\text{stress} \times \text{strain} = \frac{\text{stress}^2}{2Y}$$

Wire as spring: $k = \dfrac{YA}{L}$; series/parallel like springs

Elongation under own weight: $\Delta L = \dfrac{MgL}{2AY} = \dfrac{\rho g L^2}{2Y}$

Thermal stress (ends fixed): $F/A = Y\alpha\,\Delta T$

Max length hanging wire: $L_{\max} = \dfrac{\text{breaking stress}}{\rho g}$

Bulk compression: $\dfrac{\Delta V}{V} = \dfrac{\Delta P}{B}$; $\dfrac{\Delta\rho}{\rho} = \dfrac{\Delta P}{B}$

⋆ Common Pitfall

- Work done by load $= Mg\Delta L$, but energy stored $= \frac{1}{2}Mg\Delta L$; half is lost as heat
- Breaking force $\propto A$; breaking stress is a material constant

4. FLUID PRESSURE & BUOYANCY

$$P = P_0 + \rho g h, \qquad P_{\text{gauge}} = P - P_0$$

$P_0 = 1$ atm $= 1.013 \times 10^5$ Pa $= 76$ cm of Hg

Pascal's law (hydraulic lift): $\dfrac{F_1}{A_1} = \dfrac{F_2}{A_2}$

Archimedes: buoyant force $F_B = \rho_\ell V_{\text{sub}}\, g$

Apparent weight: $W' = W\left(1 - \dfrac{\rho_\ell}{\rho_b}\right)$

Floating: $\dfrac{V_{\text{sub}}}{V} = \dfrac{\rho_b}{\rho_\ell}$

Accelerating container: $\tan\theta = \dfrac{a}{g}$ (free surface tilt)

Rotating liquid: rise at radius r: $h = \dfrac{\omega^2 r^2}{2g}$

Lift accelerating up: $P = P_0 + \rho(g+a)h$

⋆ Common Pitfall

- Pressure depends on depth, not on container shape (hydrostatic paradox)
- Buoyancy acts at centre of mass of displaced liquid (centre of buoyancy)
- In free fall, buoyancy $= 0$

5. FLUID DYNAMICS

Continuity: $A_1 v_1 = A_2 v_2$ (volume flux $Q = Av$)

Bernoulli:

$$P + \tfrac{1}{2}\rho v^2 + \rho g h = \text{constant}$$

Torricelli (efflux): $v = \sqrt{2gh}$ (h below free surface)

Hole at depth h in tank of height H:
Range $R = 2\sqrt{h(H-h)}$, max at $h = \frac{H}{2} \Rightarrow R_{\max} = H$

Time to empty: $t = \dfrac{A}{a}\sqrt{\dfrac{2H}{g}}$

Venturimeter: $v_1 = a\sqrt{\dfrac{2\Delta P}{\rho(A^2 - a^2)}}$

⋆ Common Pitfall

- Bernoulli: steady, incompressible, non-viscous, streamline flow, along a streamline
- Faster flow $\Rightarrow$ lower pressure (lift on wings, roofs blown off, spinning ball)
- Holes at h and $(H-h)$ give the same range

6. VISCOSITY

$$F = -\eta A\frac{dv}{dx}, \qquad [\eta] = [ML^{-1}T^{-1}], \quad \text{SI: Pa}\cdot\text{s}$$

Stokes: $F = 6\pi\eta r v$

Terminal velocity:

$$v_t = \frac{2r^2(\rho-\sigma)g}{9\eta}$$

Reynolds number: $R_e = \dfrac{\rho v D}{\eta}$; < 1000 laminar, > 2000 turbulent

Poiseuille: $Q = \dfrac{\pi P r^4}{8\eta l}$

⋆ Common Pitfall

- $v_t \propto r^2$; if $\rho < \sigma$ (air bubble in liquid), v_t is upward
- Viscosity of liquids decreases with T; of gases increases with T
- 1 poise = 0.1 Pa·s

7. SURFACE TENSION

$$S = \frac{F}{L} = \frac{\Delta W}{\Delta A} \ [\text{N/m} = \text{J/m}^2]$$

Object	Excess pressure	Surfaces
Liquid drop	$2S/R$	1
Soap bubble (in air)	$4S/R$	2
Air bubble in liquid	$2S/R$	1

Work to blow soap bubble: $W = 8\pi R^2 S$ (two surfaces)

Merging n drops: $R = n^{1/3}r$, energy released $= 4\pi r^2 S\,(n - n^{2/3})$

Capillary rise:

$$h = \frac{2S\cos\theta}{\rho g\, r} \qquad (\text{Jurin: } hr = \text{const.})$$

- $\theta < 90^\circ$ (water–glass): rise, concave meniscus
- $\theta > 90^\circ$ (mercury–glass): fall, convex meniscus
- Tube shorter than h: liquid does not overflow; R adjusts so $hR = \text{const.}$

⋆ Common Pitfall

- Soap bubble has **two** surfaces: $4S/R$ and $8\pi R^2 S$
- S decreases with temperature and with impurities like detergent
- Capillary rise is zero in free fall / satellite (liquid fills the tube)

8. JEE SHORTCUTS & MASTER TIPS

1. $\Delta L = \dfrac{FL}{AY}$; same material & load: $\Delta L \propto \dfrac{L}{r^2}$
2. Energy density $= \dfrac{\text{stress}^2}{2Y}$
3. Floating fraction $= \rho_b/\rho_\ell$
4. Efflux $v = \sqrt{2gh}$; max range $= H$ at $h = H/2$
5. $v_t \propto r^2$, so drag on a drop at $v_t \propto r^3$
6. Capillary $h \propto \dfrac{1}{r}$; bubble $\Delta P = 4S/R$

⋆ Master Confusion List

1. **Elastic** means hard to deform: steel > rubber
2. **Bulk modulus** has a minus sign; ΔV is negative under compression
3. **Drop vs bubble:** $2S/R$ vs $4S/R$
4. **Bernoulli** applies along a streamline, ideal fluid only
5. **Viscosity** vs T: liquids ↓, gases ↑
6. **Energy stored** in a stretched wire is $\frac{1}{2}F\Delta L$, not $F\Delta L$

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